# ISOMORPHISMS ARE EXACTLY THE MORPHISMS WITH ZERO CONE OBJECT

THE STACKS PROJECT AUTHORS

*Editorial note.* Counted span is the author's own proof environment, source lines 844-853, following the statement at 832-842.

*Editorial note.* No omissions: the TR3 map is displayed and both directions are executed through the named isomorphism-of-triangles lemma.

*Allowed standard prerequisites (editorial).*

- Isomorphism of triangles. lemma-third-isomorphism-triangle (carded L26221, cited at ~847). (1, f, 0) is an isomorphism of triangles in both directions.

*Source reference.* Original lemma 014A: sources/stacks-derived/pinned-original-source.tex, source lines 631–631. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

**Lemma 0.1.** *Let $\mathcal{D}$ be a pre-triangulated category. Let $f : X \to Y$ be a morphism of $\mathcal{D}$. The following are equivalent*

(1) *$f$ is an isomorphism,*
(2) *$(X, Y, 0, f, 0, 0)$ is a distinguished triangle, and*
(3) *for any distinguished triangle $(X, Y, Z, f, g, h)$ we have $Z = 0$.*

*Proof.* By TR1 the triangle $(X, X, 0, 1, 0, 0)$ is distinguished. Let $(X, Y, Z, f, g, h)$ be a distinguished triangle. By TR3 there is a map of distinguished triangles $(1, f, 0) : (X, X, 0) \to (X, Y, Z)$. If $f$ is an isomorphism, then $(1, f, 0)$ is an isomorphism of triangles by Lemma 014A and $Z = 0$. Conversely, if $Z = 0$, then $(1, f, 0)$ is an isomorphism of triangles as well, hence $f$ is an isomorphism. □

*Date*: stacks-project:derived.tex:local-snapshot-20261012; GNU FDL 1.2 or later.